Kirsten Morris

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Research Interests

Algebraic Coding Theory & Applied Discrete Mathematics, particularly Quantum Error Correction, Graph-based Codes, and Iterative Decoding

Positions Held

Postdoctoral Associate Department of Mathematics, Virginia Tech, USA — August 2025-Present

Visiting Scientist Aalto University, Finland — March 2026

NSF Mathematical Sciences Graduate Internship Fermi National Accelerator Lab — June 2022-August 2022

Education

Ph.D. Mathematics & Graduate Minor in Electrical Engineering, **University of Nebraska-Lincoln** — August 2025

- Dissertation: *Analysis of Graph-based Decoders for Quantum Low-Density Parity-Check Codes*
- Advisor: Dr. Christine Kelley

M.S. Mathematics, **University of Nebraska-Lincoln** — August 2022

M.A. Mathematics, **University of Georgia** — December 2018

B.S. Mathematics, **Georgia College & State University** — May 2016

Awards and Fellowships

Research and Travel Grants

Commonwealth Cyber Initiative, Co-PI Quantum and Post-Quantum Algorithms for Secure, High-Performance Systems, $35,000. Aug 2026-July 2027.
Embassy of Switzerland in the United States of America ThinkSwiss Summer School Scholarship, CHF 1,100. Summer 2024.

Fellowships

Research and Innovation Postdoctoral Scholar Virginia Tech, January 2026-July 2027

Graduate Fellowship for STEM Diversity National Fellowship provided 6 years of full Ph.D. funding. August 2021-May 2025

NSF Research Training Grant University of Georgia. Aug. 2016-May 2017, Summer 2017-19

Awards

Othmer Scholarship University of Nebraska-Lincoln. August 2021-August 2025

Emeritus Faculty Fellowship (awarded based on scholarship) University of Nebraska-Lincoln. November 2024

Steven Haataja Award for Outstanding Exposition by a Graduate Student University of Nebraska-Lincoln. May 2024

Publications

8. R. Freij-Hollanti, K. D. Morris, P. Šapokaitė. "Graphical Analysis of Lifted Product Code Constructions," Submitted to *IEEE Information Theory Workshop (ITW)*. arXiv:2606.11987
7. A Gómez-Fonseca, G. Matthews, K. D. Morris, T. Pllaha. "Combinatorial Analysis of Dyadic and Quasi-Dyadic Codes," Submitted. arXiv:2605.01942
6. K.D. Morris, T. Pllaha, C. A. Kelley. "Absorbing sets in quantum LDPC codes" in *Advances in Mathematics of Communications*, 2026, 23: 234-254. doi: 10.3934/amc.2026023
5. K. D. Morris, T. Pllaha, C.A. Kelley, "Effect of redundancy on syndrome-based decoding for QLDPC codes," in *Proc. of IEEE International Symposium on Topics in Coding*, 2025.
4. O. Ogunkoya, K. Morris, D. Kurkcuoglu, "Investigating Parameter Trainability in the SNAP-Displacement Protocol of a Qudit System," *Physica Scripta*, 2025.
3. J.F. Chhoa, L. Huang, A. Little, A. Maurais, K. D. Morris, M.D. van der Walt, G. Verma, and Rongrong Wang, "Supervised Dimension Reduction via Local Gradient Elongation," WiSDM 2023 Springer Proceedings *Advances in Data Science.*
2. K. D. Morris, T. Pllaha, C.A. Kelley, "Analysis of syndrome-based iterative decoder failure of QLDPC codes," in *Proc. of IEEE International Symposium on Topics in Coding*, Brest, France, 5 pages, September 2023. arXiv:2307.14532
1. G. Buhl, E. Cronk, R. Moreno, K. Morris, D. Pedroza, and J. Ryan. Matrix Completions for Linear Matrix Equations. *Involve, a Journal of Mathematics*, 10(5):781–799, 2017. 10.2140/involve.2017.10.781

Research Mentorship

Committee Member, Master's Thesis Fall 2026

Universidad Autónoma Metropolitana Unidad-Iztapalapa, Mexico

- Reviewing the thesis of Katherine Amparo Ruiz Balderrábano, titled "El qubit y códigos cuánticos correctores de errores"

Co-supervisor, Undergraduate Research Spring 2026

Virginia Tech

- Co-supervised 3 undergraduate students with Dr. Gretchen Matthews
- Project: "Analysis of absorbing sets over nonbinary fields"

Select Invited Presentations

AMS Fall Central Sectional Meeting October 2026

University of Minnesota Duluth

- Presentation: Graph-Theoretic Characterizations of Lifted Product Codes, 20 min

LIX Congreso Nacional de la Sociedad Matemática Mexicana October 2026

Universidad Autónoma de Tlaxcala, Mexico

- Presentation: Graphical Properties of Lifted Product Quantum Low-Density Parity-Check Codes, 60 min
- Session on Coding Theory, Cryptography and their Applications

27th Conference of the International Linear Algebra Society (ILAS 2026) May 2026

Virginia Tech

- Presentation: Graph Properties of Codes from Dyadic and Quasi-Dyadic Matrices, 20 min

Algebra, Number Theory, and Applications Seminar March 2026

Aalto University, Espoo, Finland

- Presentation: Graphical analysis of dyadic and quasi-dyadic codes, 45 min

Applied Algebra Seminar February 2026

Virginia Tech

- Presentation: Combinatorial properties of quasi-dyadic codes over $\mathbb{F}_2$, 45 min

Special Applied Algebra Days February 2026

University of South Florida

- Presentation: Graphical Analysis of Codes from Quasi-Dyadic Matrices, 30 min

Joint Mathematics Meeting January 2026

Washington, D.C.

- Presentation: Exploring Black-Gray-Flip for Quantum Decoding, 20 min

AMS Fall Southeastern Sectional Meeting October 2025

Tulane University

- Presentation: Redundancy Characterizations of QLDPC Codes, 20 min

International Symposium on Topics in Coding (ISTC) August 2025

UCLA

- Presentation: Effect of redundancy on syndrome-based decoding for QLDPC codes, 20 min

Applied Algebra Days June 2025

University of South Florida

- Presentation: Analysis and Improvement of QLDPC Decoding Performance, 35 min

2025 AWM Research Symposium May 2025

University of Wisconsin-Madison

- Presentation: Quantum Error Correction using QLDPC Codes, 25 minutes

Error-Correcting Codes and Combinatorial Structures Workshop March 2025

Eindhoven University of Technology, the Netherlands

- Presentation: Quantum LDPC Code Graphs & Decoding Challenges, 25 minutes

AMS Fall Central Sectional Meeting September 2024

University of Texas at San Antonio

- Presentation: Failure inducing sets in Quantum LDPC Codes, 25 minutes

ISIT Quantum Information Knowledge Workshop July 2024

Athens, Greece

- Poster: Absorbing Sets in QLDPC Codes

Virginia Tech-Swiss Coding Theory and Cryptography Workshop and Summer School July 2024

VT Steger Center, Switzerland

- Presentation: Absorbing Sets and Quantum LDPC Codes, 5 minutes

Department of Mathematics Seminar April 2024

Creighton University

- Presentation: The Art of Error: An Introduction to Coding Theory, 50 minutes

AMS Spring Eastern Sectional Meeting April 2024

Howard University

- Presentation: Absorbing Sets and Decoding Failures of QLDPC Codes, 20 minutes

Postgraduate International Coding Seminar March 2024

- Presentation: Exploring the role of absorbing sets in syndrome decoding, 50 minutes

SIAM Texas-Louisiana Annual Meeting November 2023

University of Louisiana at Lafayette

- Presentation: Graphical Characterization of Decoding Failures for Quantum LDPC Codes (25 minutes)

SIAM Southeastern Atlantic Section Annual Meeting March 2023

Virginia Tech

- Presentation: Analysis of syndrome-based iterative decoder failure of QLDPC codes (25 minutes)

Teaching Experience

Instructor of Record August 2017-May 2019, Fall 2022, Fall 2025

- Courses taught:
  - MATH 2114 Linear Algebra (Virginia Tech)
  - MATH 100A Intermediate Algebra (UNL)
  - MATH 2250 Calculus I (3x) (UGA)
  - MATH 113 Precalculus (1x) (UGA)
- Received year long classroom training in mathematics pedagogy, including theories of learning and active learning

Teaching Assistant June 2023, January 2025

University of Nebraska-Lincoln

- Math 814T: Linear Algebra for High School Teachers
- ECEN 498: Quantum Programming for STEM

STEM CONNECT Tutor August 2023-May 2024

University of Nebraska-Lincoln

- STEM CONNECT awards scholarships to and provides curricular and co-curricular support to academically talented, low-income students with interest in careers that require strong skills in mathematics or computer science
- Courses Served:
  - MATH 314 Linear Algebra (3 students)
  - MATH 221 Differential Equations (4 students)

Co-Organizer: Graduate Teaching Assistant Orientation August 2023

University of Nebraska-Lincoln

- Assisted in creating the schedule, selecting speakers, designing programming, and providing logistical support for the Math Department GTA Orientation

Professional Development

Bridging Theoretical Discrete Math Advancements with Open Source Resource Development September 2026

Banff International Research Station (BIRS), Banff, Canada

Project NExT August 2026-August 2027

- Selected for this national program administered by the Mathematical Association of America, designed to "support new PhDs in mathematics as they begin their careers."

Workshop on Applications of Commutative Algebra and Coding Theory June 2026

Centro de Investigación en Matemáticas, Guanajuato, Mexico

- Researched the weight distribution of Affine Cartesian codes.

Workshop on Combinatorial Coding Theory April 2026

American Institute of Mathematics (AIM), Pasadena, California

- Researched the construction of asymptotically good RAA codes & hierarchical LDPC codes.

VT-Swiss Coding Theory & Cryptography Summer School and Collaboration Workshop July 2024

Virginia Tech Steger Center, Riva San Vitale, Switzerland

- Researched the construction of good quantum MDS codes.

Women in the Science of Data and Machine Learning (WiSDM) Research Workshop August 2023

Institute for Pure and Applied Mathematics (IPAM)

- Investigated the efficacy of a certain distance metric for prediction of unlabeled data.

Rethinking Number Theory June 2023

AIM Research Community

- Investigated using group algebras to construct linear codes with desirable properties.

EDGE Summer Program Participant June 2016

The Enhancing Diversity in Graduate Education (EDGE) Program

Research Experience for Undergraduates (REU) Summer 2015

California State University Channel Islands

Leadership & Service

MAA Mathfest Panelist August 2026

Served as a panelist for *Advocacy: A Shared Responsibility for Our Community*

Virginia Tech Collaboration Workshop July 2026

Co-organized this international workshop with Gretchen Matthews, Hiram López, Jason LeGrow, and Rodrigo San-Jóse at the VT Steger Center in Riva San Vitale, Switzerland

JMM Special Session Co-Organizer Present-January 2027

Co-organizing an AMS Special Session on *Coding Theory: From Foundations to Frontiers* with Luca Bastioni, Giuseppe Cotardo, Rafael D'Oliveira, Andrea Di Giusto, Rodrigo San-José, Julia Shapiro

AMS Southeastern Section Special Session Co-Organizer March 2026

Co-organized an AMS Special Session on *Coding Theory and its Applications to Quantum Error-Correction and Cryptography* with Gretchen Matthews and Rodrigo San-José

JMM Special Session Co-Organizer January 2026

Co-organized an AMS Special Session on *Coding Theory Trends and Modern Applications* with Giuseppe Cotardo, Alberto Ravagnani, Ferdinando Zullo, Hiram H. Lopez, Eduardo Camps Moreno, Allison Beemer, and Rafael D'Oliveira

An Introduction to Magma Workshop October 2025

Co-organized with Giuseppe Cotardo and Julia Shapiro

Moderator September 2025

Cybersecurity & the Intelligence Community Jobs & Internship Panel Discussion

Nebraska Conference for Undergraduate Women in Mathematics (NCUWM) August 2023-May 2025

Organizing Committee Member and Volunteer

Postgraduate International Coding Theory Seminar (PICS) Co-Organizer August 2024-Present

Co-organize this online seminar for graduate students and early career postdocs with Julia Shapiro, Virginia Tech, and Andrea Di Giusto, Eindhoven University of Technology

Graduate Student Assembly August 2023-May 2025

University of Nebraska-Lincoln

- Vice President of Finance, August 2024-May 2025
- Math Department Representative, August 2023-May 2024

Math Department Graduate Mentoring Program Coordinator August 2024-May 2025

University of Nebraska-Lincoln

Discrete & Coding Theory (DISCO) Seminar Co-Organizer August 2023-May 2024

University of Nebraska-Lincoln

Graduate Student Mentor August 2023-May 2024

University of Nebraska-Lincoln

- Mentor for a first-year math graduate student. Meet weekly and on an as-needed basis to help support a successful transition to graduate school.
- Mentor for an undergraduate student interested in graduate school. Meet quarterly to discuss goals and career plans.

Panelist Oct 2022
March 2022
October 2021

University of Nebraska-Lincoln

- Math Club-"Getting into Graduate School: Applications"
- Admitted Student Weekend- "Being a Math Grad Student at UNL"
- Graduate School Workshop- "Getting into Graduate School and Applying for Fellowships"

President, Vice-President, Treasurer March 2014-May 2019

GCSU & University of Georgia
Association for Women in Mathematics Chapters

Mathematics Active Learning Team (MALT) Member Fall 2018-Spring 2019

University of Georgia

Community Outreach

Research Day Lightning Talk Fall 2026

Virginia Tech

- Gave a lightning talk introducing my research to students in the Department of Mathematics

Blacksburg Math Circle February 2026, September 2025

Virginia Tech

- Helped lead interactive enrichment activities for children

Girls Inc. Math Instructor July 2023

University of Nebraska-Lincoln

- Designed and facilitated week-long programming for middle school girls
- Led the campers through interactive mathematics activities

AGAM: Discover Cryptography Workshop Leader June 2023

University of Nebraska-Lincoln

- Designed and facilitated a workshop for high school students on "The Mathematics of QR Codes" as part of a week long math camp

EvenQuads: Notable Women in Mathematics-Data Collector February 2022

Association for Women in Mathematics

- Collected data on five notable mathematicians in pursuit of the assembly of Deck Two of Evenquads, a card game featuring woman mathematicians from the global mathematics community who have made significant contributions to the field

Athens Area Girls Math Team Coach August 2016-April 2017

STEM Pump

- Led weekly meetings with girls in grades 1-4. Used workbooks, logic games, and puzzles to introduce new mathematical concepts, develop logic and critical thinking skills, and increase the mathematical abilities and confidence of the participants

Technology

SageMath, Python, Matlab

References

Dr. Gretchen Matthews
Professor, Department of Mathematics
Virginia Tech
gmatthews@vt.edu

Dr. Christine Kelley
Aaron Douglas Professor of Mathematics
University of Nebraska-Lincoln
ckelley2@unl.edu

Dr. Tefjol Pllaha
Assistant Professor, Department of Mathematics and Statistics
University of South Florida
tpllaha@usf.edu